

2. Mission Architecture & Concept of Operations

NASA and Sauder et al [1, 2] performed studies on mission architecture, concluding that a flagship orbiter such as VERITAS is the best supporting architecture for the rover's mission. A small satellite constellation was seen as a potential alternative. However, these studies did not provide a detailed assessment of technical architecture, associated risks, and concept of operations involving the group has successfully achieved the primary objectives set at the start of the project. We developed a fully mechanical rover, designed to traverse the Venus terrain autonomously. The final design features inherited, redesigned, and new subsystems integrated with each other to form a complete responsive system. The group strictly adhered to the design process set out at the start, we used 3D printing, laser cutting, and water jet cutting for rapid prototyping to optimise project resource usage and design breadth. The group successfully tested the rover and fulfilled all design requirements for each subsystem, in addition to completing these primary objectives the team also developed a full rover and end-to-end radar simulation to further validate our subsystem designs. Shown in figure 9. The mission architecture was determined according to key scientific themes identified by the Science Division of the Mission Directorate, which are described in the Venus – A Scientific Review.

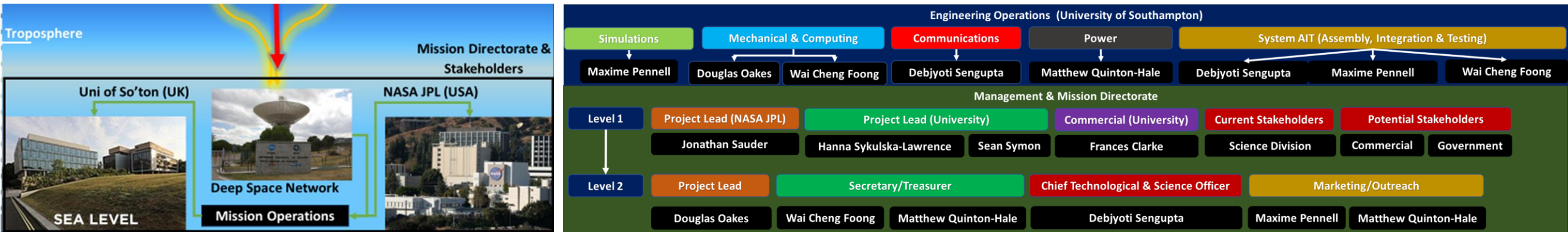


Figure 9: Mission Directorate for VESPER Rover Mission

NASA’s Deep Space Network team is responsible for mission data handling, which is further processed by NASA JPL and the University, who share it with relevant stakeholders. The data on resources found on the Venusian terrain is of interest to commercial organizations and government agencies, while data on the rover's performance helps stakeholders introduce innovations and optimize design performance and reliability. The key to successful mission concepts and robust analytical studies lies in two core principles - one must get their initial assumptions verified since they tend to carry over during the implementation stage, and the decisions made during this phase have far-reaching consequences throughout the course of the project and the mission. Figure 10 allow us to understand the mission design which forms the framework of the System Requirements Matrix and the Technical Design Brief.

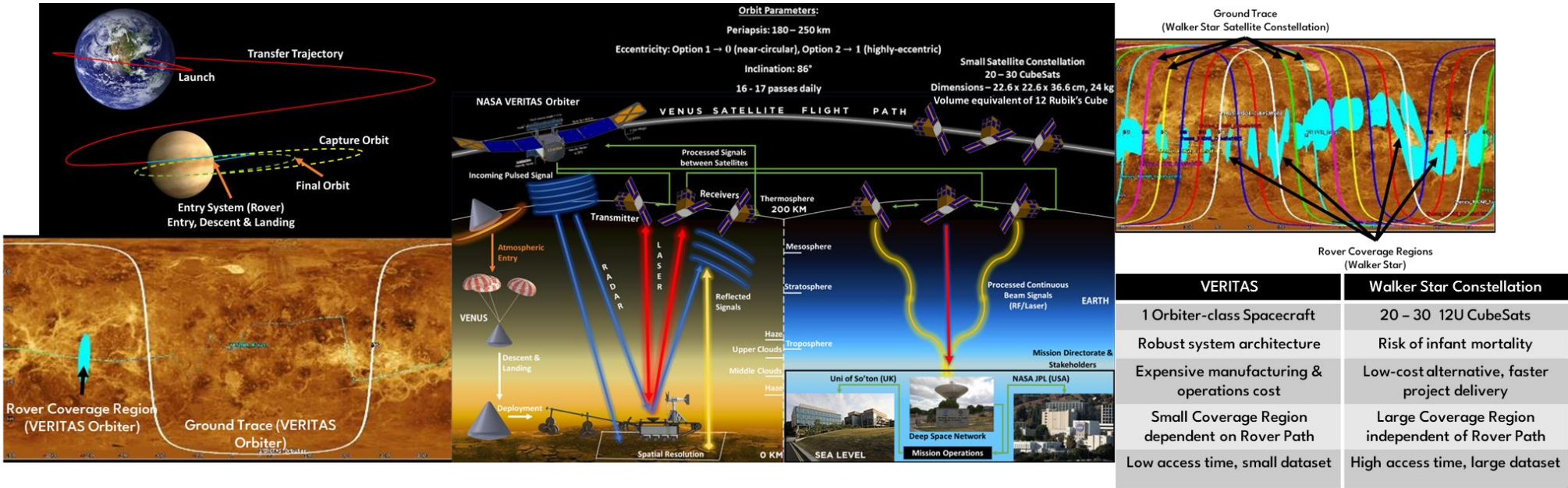


Figure 10: Mission Architecture & Concept of Operations for VESPER Rover Mission compared with NASA VERITAS Orbiter and proposed 86 deg : 30/10/1 Walker Star Small Satellite Constellation

The mission commences with the launch and transit to Venus, where the Orbiter-Entry System (O – ES) package enters an initial capture orbit. The orbiter then adjusts its orbit by performing a retrograde burn to enter a desired circular or elliptical orbit while releasing the ES capsule into Venusian atmosphere. After surviving atmospheric entry lasting an hour, the capsule deploys parachutes to slow down its descent. Upon landing, the VESPER rover would be deployed. Mission data collected through the reflected signal would be transmitted back to ground stations such as NASA-DSN and shared with stakeholders for various applications. The comparative mission analysis was conducted using Systems Tool Kit (STK) simulation software with factors such as raw data collected, solar power, and access time with respect to the rover determining the suitable architecture, the results of which have been tabulated in Appendix B. The team recommends the use of a small satellite constellation contrary to the recommendations by Sauder et al [2]. While the flagship architecture provides a robust system with minimum complexities, the high development and operations costs make the small satellite constellation, consisting of 12U CubeSats, a viable alternative. The constellation architecture enables collection of over 150 times the raw data collected by VERITAS due to extended access time, with the coverage area being 28 times larger than VERITAS. However, these gains come with the technical challenges of rigid power management constraints and the risk of infant mortality (early satellite failures during the mission), necessitating redundancy.

4. Design Process

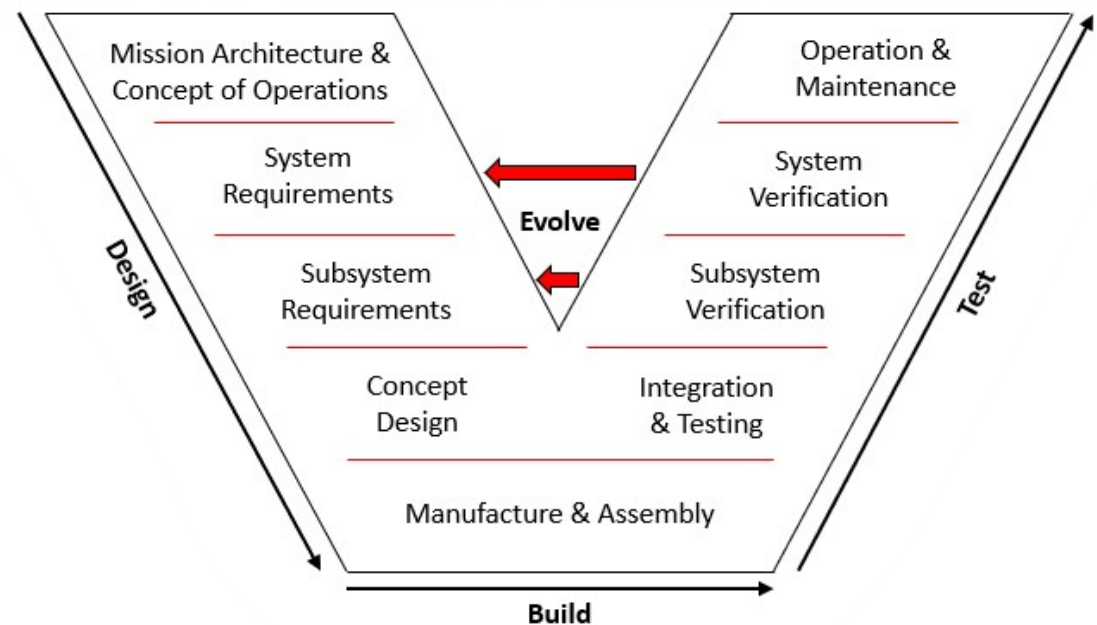


Figure 14: Systems Engineering Design Process for VESPER

The design process of the VESPER project, illustrated in Figure 14 is the cornerstone of the team’s iterative approach to the design of the current Rover. Taking inspiration from the system’s engineering V-diagram, the team’s design process is broken down into 4 sections: design, build, test, and evolve. From scope definition and decomposition of the project objectives, the team has been able to construct system and subsystem-level requirements to design the VESPER Rover against, ensuring concepts provide suitable solutions to the problems at hand.

Utilising the Design for Manufacture approach, designs have been condensed and refined to maximise the use of 3D printed and laser cut parts. This, in addition to the continuous use of rapid prototyping, has enabled the team in multiple instances, to design and test several concepts addressing a single problem, trading-off between the concepts to finalise the most effective solution. Emphasis of the design has also been placed on ensuring subsystems are modular. This is especially key for a project that will see several iterations such as our own, as the redesign and refinement of subsystems should not cause interfacing subsystems to be unnecessarily redesigned.

Using phased component, subsystem and integration testing, the team has conceptually proven all design aspects against the requirements set within the project scope. From individual mechanisms and working principles to subsystem interfaces and integration, the team has utilised a variety of university facilities and self-designed testbeds to record and document subsystem performance. The team has also identified areas of further refinement of the developed subsystems that could be implemented in future work to allow designs to meet the outlined requirements more effectively, improving system reliability and efficiency.

With the VESPER project being in its second year of development, the Evolve stage of the process has been crucial to delivering the best possible system design. By testing last year’s subsystem proof of concepts, the team has been able to document salient features of the existing designs which both exceed and fall short of the defined requirements. By utilising the insight gained, the existing designs have been adapted and evolved to overcome the identified shortcomings and produce a refined subsystem and system design that now meet the project requirements. The evolve stage is also crucial moving forward with the project, allowing teams to build upon developed subsystems and further refine identified areas of improvement, allowing the VESPER project to become iterative, constantly utilising and improving upon previous iterations of engineering design.

5. System Design

5.1. System Requirements Matrix

Top-level requirements of the system shown in Table 3 are derived from the requirements of the ‘Exploring Hell’ challenge and AREE reports, to align the project output with the interest of the Mission Directorate. The Mission Directorate comprises of a diverse range of stakeholders from academia and government to technology and commercial partners.

Level 1 (L1) Primary Requirements	Level 2 (L2) Derivative of L1 Requirements	Level 3 Derivative of L2 Requirements
L1-1: VESPER shall make science observations over at least 114 days.	L2-1: VESPER shall traverse a minimum of 35 km (extended to 100 km), traverse wrinkle ridges such as the tesserae, and built robust enough to survive 114 days	L3-1: VESPER shall produce at least 40 W of Power/day for mobility
		L3-2: VESPER shall have survival power to 0.3 m/s. Nominal mission case shall be designed around 0.6 m/s
		L3-3: VESPER shall be able to ascend or descend slopes of at least 35 degrees with obstacles up to 50 cm high and 200 cm wide
		L3-4: VESPER shall be able to drive against a wind speed of 2.2 m/s
		L3-5: VESPER shall be designed to withstand corrosion, operate in 482°C and 90 bar for 114 days
L1-2: VESPER shall not require future developments in high temperature electronics to achieve minimum requirements.	L2-2: VESPER shall transmit data without the development of high temperature electronics	L3-6: VESPER shall be designed so it is not inoperable if it flips, or so that it will not flip.
		L3-7: VESPER shall transmit data at a minimum rate of 50 bits/s with VERITAS without electronics
	L2-3: VESPER shall fit within the constraints of existing EDL systems	L3-8: VESPER shall fit in an aeroshell of 3.2 metres in diameter
		L3-9: VESPER shall have a payload mass of less than 1100 kg, including margin
	L2-4: VESPER shall navigate the Venus terrain without modern electronics	L3-10: VESPER shall take landing loads without damage.
		L3-11: VESPER shall sense and circumnavigate obstacles it cannot overcome

Table 3: VESPER Mechanical Rover System Requirement Matrix

5.2. System Design Overview

The current VESPER mechanical Rover comprises 6 main subsystems that enable the Rover to meet the system requirements set for the scope of the project. For the Rover to sequentially perform its navigation of terrain, these subsystems are the obstacle detection subsystem, the mechanical computer, the output computer, the drive train, the locomotion subsystem, the power subsystem, and the communications subsystem. These subsystems are mounted onto a chassis which serves as the structural support and suspension for the entire Rover, while allowing the subsystems to be integrated with one another through a common structure. The system design overview is shown in Figure 15 with each subsystem described in the text boxes.

Obstacle Detection Subsystem

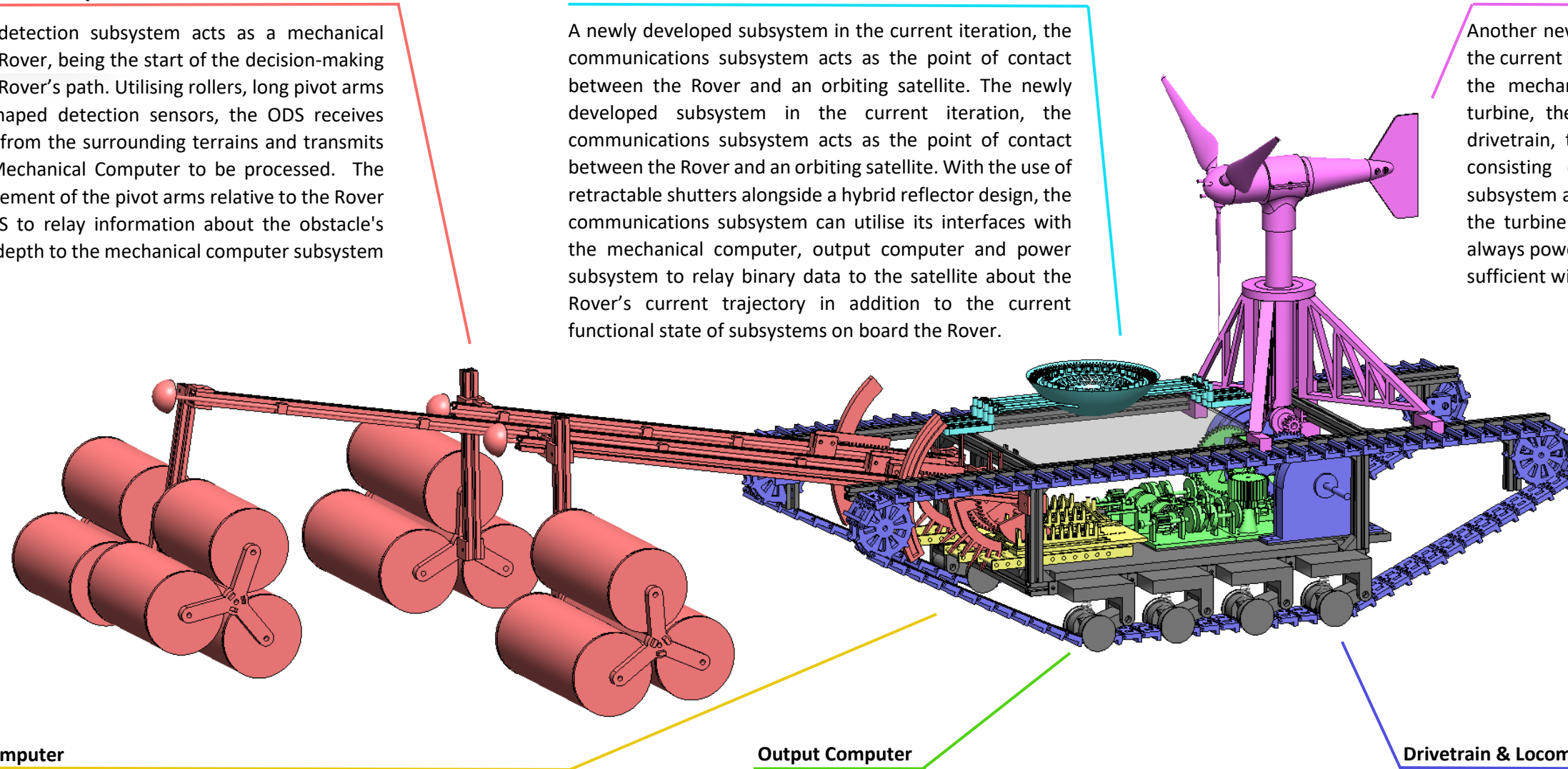
The obstacle detection subsystem acts as a mechanical sensor for the Rover, being the start of the decision-making process of the Rover's path. Utilising rollers, long pivot arms and plunger-shaped detection sensors, the ODS receives various inputs from the surrounding terrains and transmits them to the Mechanical Computer to be processed. The angular displacement of the pivot arms relative to the Rover allows the ODS to relay information about the obstacle's height and/or depth to the mechanical computer subsystem for processing.

Communications Subsystem

A newly developed subsystem in the current iteration, the communications subsystem acts as the point of contact between the Rover and an orbiting satellite. The newly developed subsystem in the current iteration, the communications subsystem acts as the point of contact between the Rover and an orbiting satellite. With the use of retractable shutters alongside a hybrid reflector design, the communications subsystem can utilise its interfaces with the mechanical computer, output computer and power subsystem to relay binary data to the satellite about the Rover's current trajectory in addition to the current functional state of subsystems on board the Rover.

Power Subsystem

Another new subsystem integrated onto the Rover in the current iteration, the power subsystem gives life to the mechanical Rover. Utilising a horizontal wind turbine, the power subsystem provides power the drivetrain, through its interface with the drivetrain consisting of bevel and spur gears. The power subsystem also features a yaw mechanism that allows the turbine to face any oncoming wind direction to always power the rotation of the drive train whenever sufficient wind speeds are present.



Mechanical Computer

The mechanical computer (MC) is the decision-making subsystem onboard the Rover. By interfacing with the obstacle detection subsystem, where information regarding an obstacle's size and position relative to the Rover is gained, the mechanical computer can determine if the detected obstacle is passable or not. If the obstacle is deemed impassible, the mechanical computer will communicate with the output computer, through an interface between the two subsystems. The resulting decision determines which direction the Rover should reverse and turn to, to avoid and navigate around the obstacle.

Output Computer

The output computer subsystem acts as the interface between the mechanical computer which decides the appropriate manoeuvre the Rover should perform, and the drivetrain which actuates the manoeuvre to make that decision a reality. Taking in an input from the mechanical computer and actioning it as an output to the drivetrain, the output computer also utilises constant integration with the power subsystem to ensure a manoeuvre is carried out for the full pre-determined duration of time before the Rover returns to its original forward motion state.

Drivetrain & Locomotion Subsystem

The drivetrain and locomotion control the motion of the VESPER Rover. When the interface with the output computer is actuated, the drivetrain uses a planetary gear mechanism to switch the left and right sides of the Rover from forward drive, to reverse motion, independently of the other side. The drive train then outputs this resultant motion, for each side of the drive train, to the locomotion subsystem's sprockets which are used to drive the caterpillar tracks and the overall motion of the Rover system.

Figure 15 - Diagram depicting the location of subsystems in the final Rover design and their basic functions

6.6. Communications Subsystem

The Communications subsystem (Figure 79), one of two new subsystems developed this year, is tasked with transmission of data collected during the Rover’s mission on the surface while assisting with positional tracking. Being a mission-critical subsystem, conventional silicon-based electronics cannot survive under Venusian conditions. Considering the long duration of the mission, this warrants the team to develop an alternative solution without using silicon carbide-based high temperature electronics based on principles of passive communications (e.g., reflection).

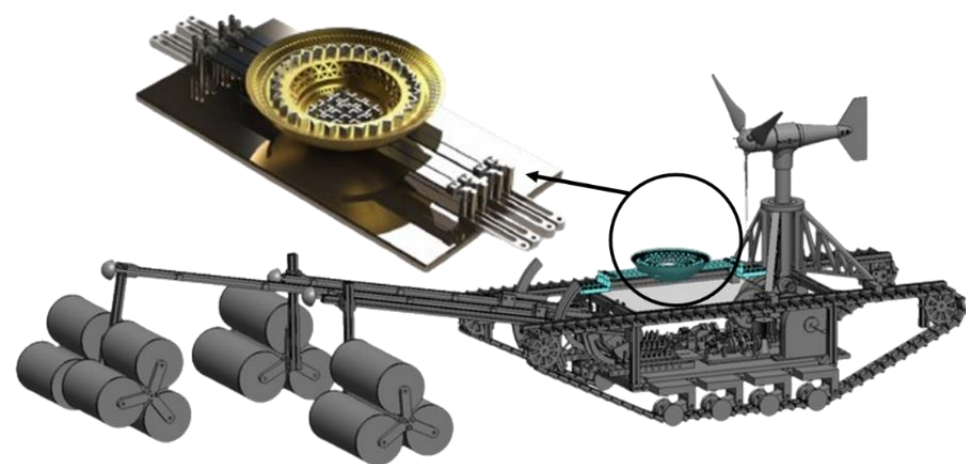


Figure 79: Location of the communications subsystem in VESPER Rover (highlighted in cyan), alongside real-life render of the subsystem.

A. System Requirements

Requirements	Justification	Validation
COMM1: The Communications subsystem must be able to transmit data and communicate with support architecture mechanically.	System Requirements Matrix L1-2: VESPER shall not require future developments in high temperature electronics.	Testing: Integrated testing with subsystem and test bed connected to simulated onboard satellite computer (e.g., Arduino)
COMM2: The Communications subsystem must be able to withstand Venusian environment	L3-5: VESPER shall be designed to withstand corrosion, operate in 482°C and 90 bar for 114 days.	Literature Review: Referencing materials research done by NASA and Ansys Granta Edupack
COMM3: The Communications subsystem must be able to reflect pulsed and/or continuous signals of radio waves and/or laser while being a compact design	L2-2: VESPER shall transmit data without the development of high temperature electronics. L3-8: VESPER shall fit in an aeroshell of 3.2 metres in diameter	Simulation & Testing: Integrated testing with subsystem and test bed connected to simulated onboard satellite computer (e.g., Arduino), comparison with simulation results
COMM4: The Communications subsystem must be able to transmit scientific data and system telemetry using suitable protocol at an acceptable bit rate	L2-2: VESPER shall transmit data without the development of high temperature electronics, L3-7: VESPER shall transmit data at a minimum rate of 50 bits/s with VERITAS without electronics	Simulation: Validation of design against MATLAB and Ansys Systems Tool Kit (STK) simulations.

Table 11: System Requirement & Validation Table for the VESPER Communications Subsystem

B. Detectability Factors – SNR and RADAR Cross Section

The transmitter sends either continuous or pulsed beams of radio waves or light, depending on the communication architecture chosen in the Mission Architecture & Concept of Operations. Due to the large wavelength of radio waves, radar-based design constraints were chosen but this design choice makes the system adaptive for laser communications as well.

The carrier waves are directed towards a target – stationary or moving, which has its distinct reflectivity depending on the materials from a range, R . As a result, the reflected beam of radio waves or laser has different signal characteristics in comparison to the incident carrier waves. The change in these characteristics allows us to determine the state of the target and process information transmitted by the target with the help of a computing unit. To effectively communicate with the target, the incident transmission power, P_t must be greater than the reflected power, P_r to account for system energy losses, L , and losses due to the environment, P_n . The ratio of the incident power to the power of the reflected waves from the target is referred to as the signal-to-noise ratio (SNR), given by Eq. 6.6.1. SNR values are generally greater than 1 for greater, precise, and accurate detectability since any value below 1 lead to lack of detection due to the signal power not enough to be distinguished from the background noise.

$$SNR = \frac{P_t}{P_r} = \frac{P_t G_t^2 \lambda^2 \sigma}{(4\pi)^3 R^4 P_n L}, \quad G_t = \text{transmitter gain}$$

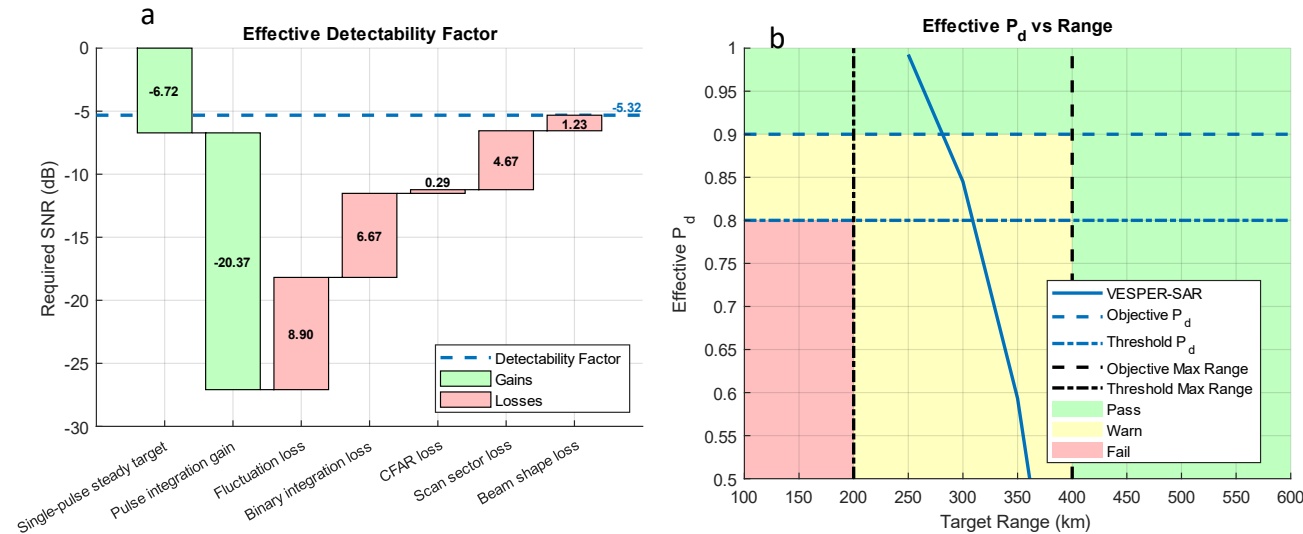
6.6.1

Not only can Eq. **Error! Reference source not found.** can be used to determine the required SNR for effective detection, but it is also able to determine the required aperture, also known as the RADAR Cross Section (RCS). From Eq. **Error! Reference source not found.**, we can determine that the RCS of a communications system is proportional to the fourth power of range R while having an inverse relation to the square of the wavelength of the EM wave. Computation of design constraints for the system was done using MATLAB in tandem with the Iridis supercomputing clusters. The effective SNR and associated probability of detection for target range P_d was calculated after computation of signal processing, searching, and tracking losses for the communication system, as shown in Figures 80 and Figure 81. This was done alongside computation of free-space propagation and atmospheric losses using MATLAB toolboxes and our new medium-fidelity systems simulation and modelling of the rover. For the transmitter antenna used by VERITAS and Walker Star Constellation discussed in Mission Architecture & Concept of Operations., given by table 11, the required SNR was calculated to be 57dB (see 6.6.3), consistent with JPL data [2].

Parameter	VERITAS Value	Satellite Constellation
Power + Gain	260 W + 25 dB transmitter gain	20 W + 33 dB transmitter gain
Wavelength	3.8 cm (8 GHz Radio Wave)	3.8 cm (8 GHz Radio Wave)
Antenna Dimensions	1.8 m diameter	1.8 m
Bandwidth	20 MHz	20 MHz
Pulse Length	35 us	35 us
Pulse Repetition (Sampling) Frequency	5600 Hz	13085 Hz
Height	240 km	240 km
System Noise Temp	1226 K	1226 K
Surface Beam Vel (Orbital Velocity)	6.9 km/s	7.15 km/s

Table 10: Transmitter Antenna Configuration for VERITAS & Walker Star Small Satellite Constellation

Figure 82: Target RADAR Cross-Section (RCS) Calculation using Radar Equation Calculator in MATLAB



Figures 80 (a) and (b): (a) System loss bar chart for given RADAR configuration, (b) Probability of detection curve against target range

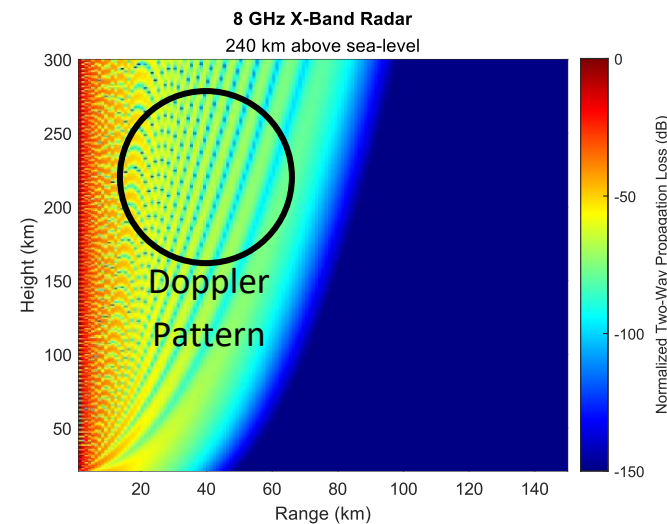


Figure 81: Normalised Two-Way Propagation Loss for 8 GHz RADAR Signal accounting for free-space and atmospheric losses

$$\text{Required SNR} = \text{One - Way Propagation Loss} + \text{System Loss} + \text{Atmospheric Loss} - \text{Transmitter Gain} \quad 6.6.2$$

$$\text{Required SNR} = 80.2031 - 5.32 + 7 - 33 = 56.8831 \text{ dB} \quad 6.6.3$$

The associated RCS for the given SNR was identified at 20 dBsm (decibel per square metre), thus giving us one of the key design constraints towards the design of the Subsystem Design.

Radar Specifications		Peak Transmit Power:	
Wavelength:	3.8 m	260	W
Pulse Width:	35 μs	SNR:	16.2718 dB
System Losses:	-5.32 dB	Detection Specifications for SNR	
Noise Temperature:	1226 K	Probability of Detection:	0.99
Target Radar Cross Section:	20 dBsm	Probability of False Alarm:	1e-3
Configuration:	Monostatic	Number of Pulses:	50
Gain:	25 dB	Swerling Case Number:	1
		Target Range:	236.8 km

C. Subsystem Design

Reflector Target

The reflector target design employs similar design constraints as the design of an antenna. Multiple concepts of the reflector target were explored based on its shape, reflection and scattering properties, and ability to meet the target range. As part of the simulations conducted to determine Detectability Factors – SNR and RADAR Cross Section, the required antenna beamwidth was computed at 1.1928° and 21.4711°. With two key design constraints being the antenna beamwidth and the required RCS (20 dBsm – shown in Figure 82) to achieve the desired SNR alongside the challenge of reflecting the transmitted carrier signal to the source, the concepts that were reviewed have been tabulated in table 12.

Concept & Key Features	
Parabolic Reflector – High gain, simple construction, precise directivity, robust for space applications, limited bandwidth, mass constraints, poor range	Cylindrical Reflector – High signal polarisation and scattering, complex construction, narrow beamwidth, suitable for scanning, limited gain
Corner Reflector – complex construction, mitigates interference and polarisation, suitable for ranging, wide bandwidth, limited gain, poor coverage	Spherical Reflector – Simple construction, reduction in signal aberration, omnidirectional coverage, poor coverage and range

Table 12: Reflector concepts and key features including strengths and weaknesses.

Since the application for the reflector target is passive communications, parameters such as high gain, precise directivity, wide bandwidth, and good range were used to determine the suitable concept. Through RCS simulations using MATLAB toolboxes run on the Iridis supercomputing cluster, it was determined that a hybrid solution of parabolic and corner reflectors is the most appropriate design. The hybrid design employs a parabolic reflector as the base, with slots where tiny corner cube reflectors would be integrated, as shown in Figure 83 . Further simulations led to the innovation of introducing corner tetrahedral cuts as focal exciters, improving the gain of the system. This innovation was inspired from the tetrahedral wedges in an anechoic chamber for noise absorption. In this case, the embedded tetrahedral cuts filters the noise but amplifies the signal enough for the system to be detected.

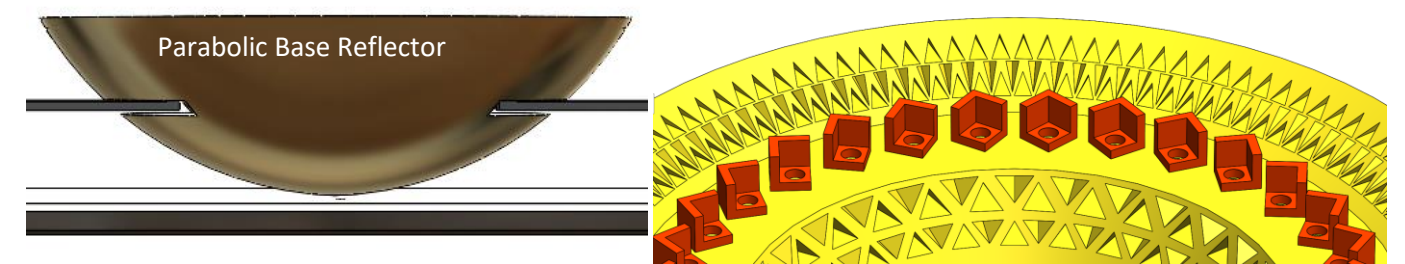


Figure 83: The design of the hybrid reflector – parabolic base (side view, left), corner tetrahedral cuts into the base (highlighted in yellow) and corner cube reflectors (highlighted in red) (right).

The performance of the system was compared against a conventional Cassegrain antenna of the same dimensions (25 cm diameter – aperture of parabolic reflector) used for space applications and radio astronomy. The hybrid system outperformed the conventional design in terms of RCS by a factor of 1.3 (shown in Figure 84), while meeting the required RCS constraint of 20 dbSm, with the RCS of the design rated at 368.7 dbSm. This design can demonstrate that the mechanical design without the use of any electronics (silicon or silicon carbide-based) is far superior to conventional antennas in terms of mass and communication performance. The elevation sweeps are also able to validate the hybrid reflector design against the beamwidth requirement while ensuring broad coverage with RCS

values of approximately 75 dbSm. This ensures that the rover and the communications system can be detected with ease by a satellite despite Venus's thick atmosphere.

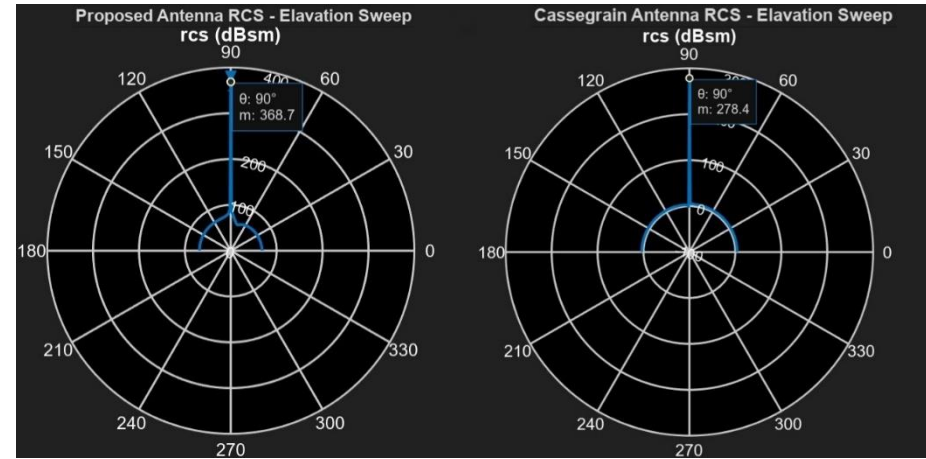


Figure 84: Radar Cross-Section Calculation using Elevation sweep, simulating a satellite passing over the Communications Subsystem and its equivalent of the Cassegrain Antenna using MATLAB-STK interface.

Corner Cube Reflector Matrix & Transmission Mode

Modern communication system designs involve the integration of onboard computing processors and data storage microchips for computation and post-processing alongside transceivers. The base of the parabolic reflector has been designed to incorporate a 5 x 5 matrix of corner cube reflectors, resembling the cores of a computer's central processing unit (CPU), as represented in Figure 85. This allows the subsystem transmit 8-bit binary data to the satellites. Each corner cube reflector has been allocated for a specific purpose to ensure a successful detection (detection core or DC) by the satellite, followed by successful data transmission (transmission core or TC), allow the subsystem transmit 8-bit binary data to the satellites shown in Figure 86. The matrix allows redundancy not only for detection but also for data transmission, as the matrix design affects the design configuration of the shutter mechanism.

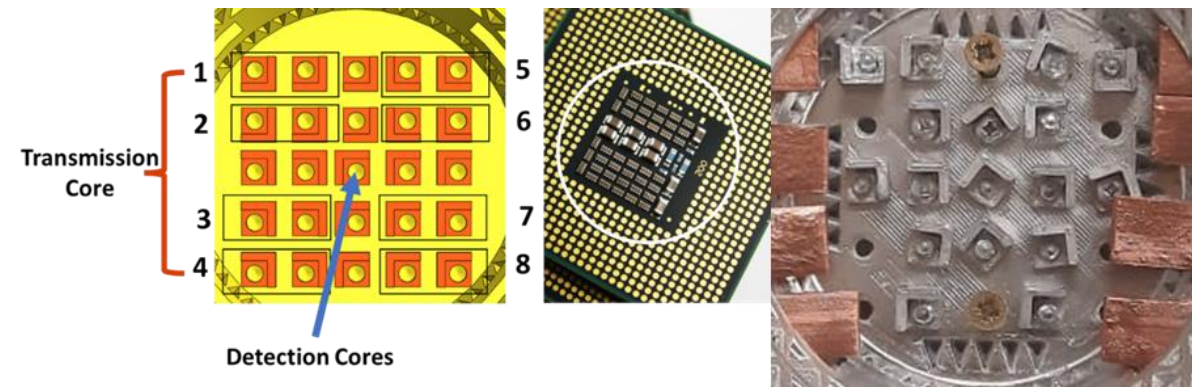


Figure 85: CAD Render of the corner cube reflector matrix compared with a CPU core matrix and image of the matrix from the prototype.

During the search mode, the satellite will specifically use the detection cores to improve alignment. Once the communications target alongside the rover is detected, the satellite will change to the track and transmit mode. In this mode, the satellite would use the detection core to track the rover throughout the access duration while the transmission cores will transmit binary data by reflecting the carrier signal. The reflected power of the carrier signal is dependent on the materials used for the corner reflector, the parabolic target, and the shutters. Once transmission is complete, the onboard satellite computer would process the raw data consisting of 1's and 0's received in the

$$\text{Raw Data Binary TC Matrix} = \begin{bmatrix} TC-1 & TC-5 \\ TC-2 & TC-6 \\ TC-3 & TC-7 \\ TC-4 & TC-8 \end{bmatrix}, \quad \text{Binary - decimal conversion} = \sum_{i=1, j=1}^n 2^{TC_{ij}} \quad 6.6.4$$

format shown in Eq. 6.6.4 **Error! Reference source not found.** with the help of suitable machine learning algorithms for RCS classification .

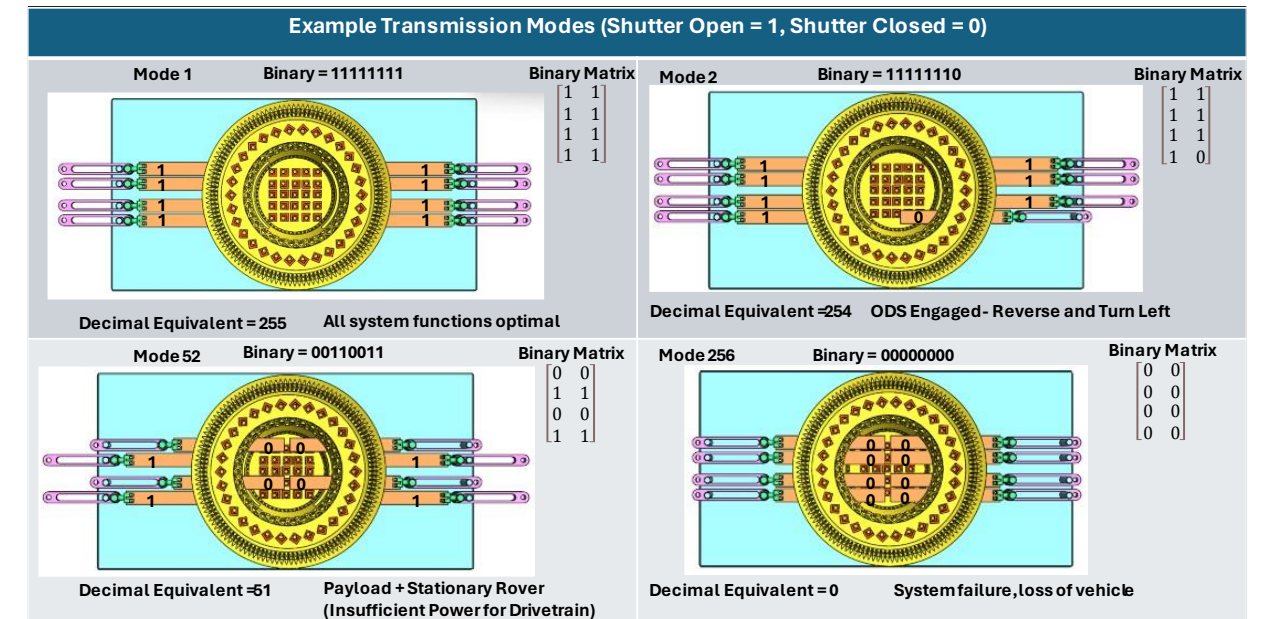


Figure 86: Example transmission modes demonstrating communication logic, alongside their respective Binary TC matrices

Shutter Mechanism, Actuators and Material Selection

To ensure the subsystem design is consistent with some of the rover's subsystems such as the Mechanical Computer Subsystem, pin-slot actuators shown in Figure 87 were used as integration interfaces with different subsystems of the rovers. These actuators allow integration of modular rods & components connected with the output channel of different subsystems. Also, for the transmission logic shown in **Error! Reference source not found.** to work, material reflectivity properties have to be different since the change in aperture area is not enough for the onboard satellite computer to process and distinguish the reflected signal. Material analysis using Granta Edupack has recommended the use of gold for the reflector target, molybdenum for the corner cube reflectors, vanadium for the shutter and nickel superalloys for the support plate on to which the subsystem is assembled. These materials were chosen to conform with the COMM2 requirement due to their corrosion resistance to forming sulphates and sulphides, durability under Venusian conditions, and manufacturability.



Figure 87: Pin-slot actuator connected with shutter and comparison with proof-of-concept model.

D. Final Design

Figure 88 is representation of the final design of the communications subsystem for VESPER Mechanical Rover.

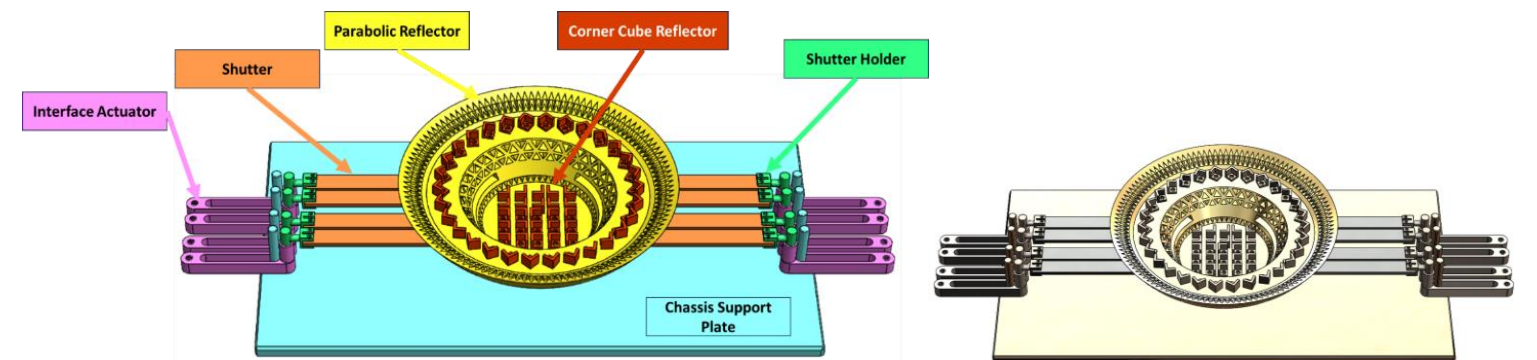


Figure 88: Annotated assembly visualisation of the Communication subsystem, alongside CAD render of the actual

7.Simulation and Modelling

The rover performance requirements set out earlier in the design process, are difficult to validate against the proposed rover design with just hand calculations. To test the rover performance in more complex scenarios, such as traversing dynamic terrain across the Venus surface, or navigating complex obstacles, a high-fidelity simulation model of the rover is needed. The model is made of a collection of subsystems that mirror the logic of systems on the proposed rover design, these models interface together to form a discrete simulation of the rover dynamics, the ODS, MC, gearbox, and locomotion systems. The communications link between the rover and satellite constellation was simulated to validate that the proposed radar system can be integrated with rover and fulfil the mission communication requirements.

Simulation Requirements	Justification	Verification
SIM1: Accurate representation of physical rover mechanics and movement.	To analyse the aspects of rover performance and accurate model of the rover needs to be made with realistic physics and terrain interaction.	Design: The physical rover model is made as a MATLAB Simscape system. This tool allows us to accurately model complex physical systems and objects.
SIM2: ODS, MC logic and their implementation should be identical to the proposed design.	Required to test how the proposed ODS integration on the rover will perform and to inform further development.	Design: Logic gates from the mechanical computer are modelled in Simulink, and physical ODS arm angles from Simscape are inputted.
SIM3: Design and test multiple terrain scenarios for the rover to traverse with a visual interface.	The rover design needs to be tested across various terrain it may face on the Venus surface.	Design and Test: Scans of the actual Venus surface and specific obstacles are used in the rover model.
SIM4: Accurate atmosphere model.	Accurate atmosphere model needed to simulate wind and RF signal losses through the atmosphere.	Design: The NASA Global Reference Atmospheric Model (GRAM) Suite is used to model Venus atmosphere.
SIM5: Simulation of proposed RF pulse radar communication system.	Communications system needs to be simulated using a high-fidelity radar model to test and prove its viability for this rover design.	Design: Several MATLAB packages are used in conjunction to model the radar system accurately.
SIM6: Rover sim data should be logged and model performance optimised.	Data and visuals for this model need to be saved and organised such that runs can be compared and analysed.	Design and Test: Data is exported from the rover model into the MATLAB Workspace.

Table 13: Requirement Table for VESPER’s MATLAB/Simulink Model

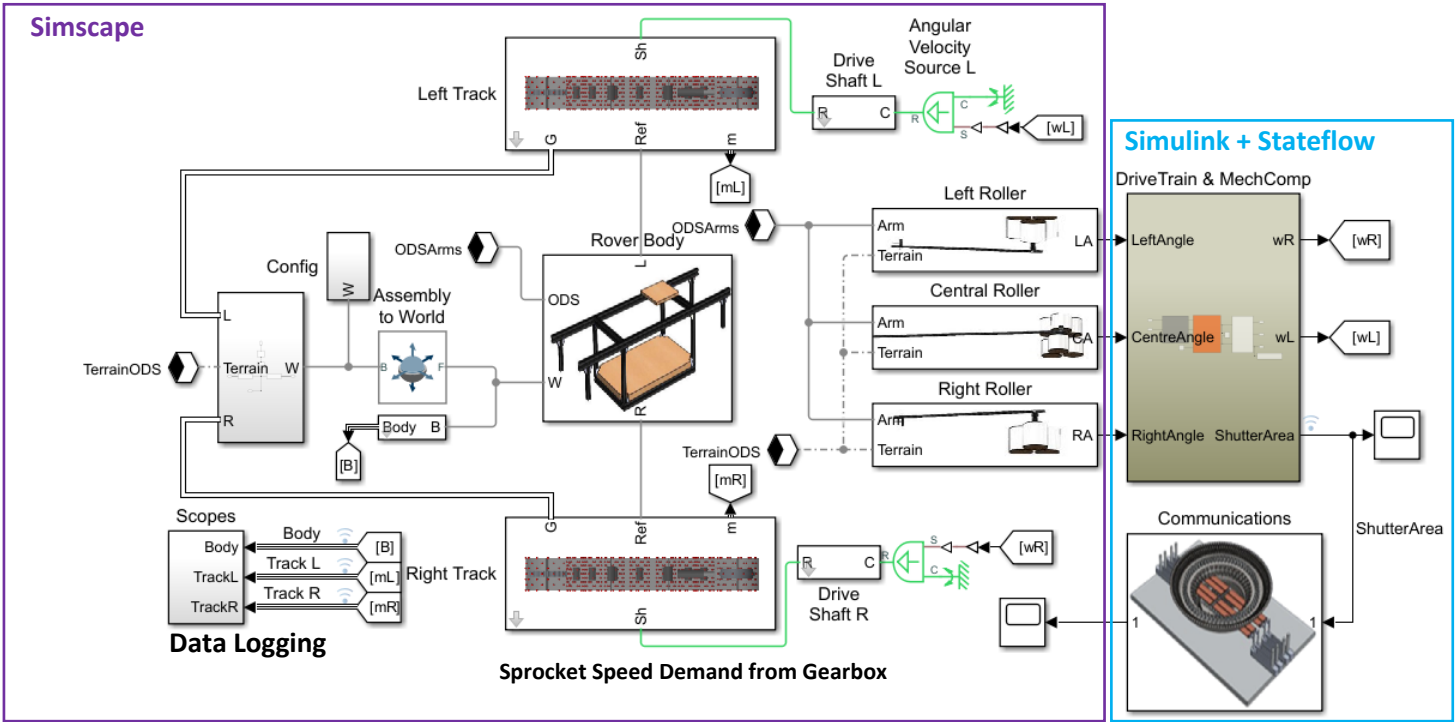
A. Literature Review and Software

The base of our rover model is developed in MATLAB. MATLAB is a programming and numeric computation platform that provides a large selection of tools used to analyse data and develop models. MATLAB is used to run the model, set simulation parameters, and organize logged data. The model is constructed using Simulink, an addon to MATLAB that enables a block diagram environment for Model-Based design. Simulink makes the simulation of complex multi domain systems and the tracking of data in the model more visually intuitive. Addons packages provide additional

tools and simulation capacity for specific purposes. Simulink interfaces directly with MATLAB and with the addons listed in Appendix C – MATLAB Toolboxes for Simulation Model to make a fully integrated simulation all within the MATLAB environment.

B. Model Structure

The Model is split into two distinct sections, Simscape handles all the rover movement, dynamics, and modelling. Each component of the rover is modelled individually and joined; this unit is then treated as a moveable 6-DOF joint on the surface of the terrain. Simulink and Stateflow are used to manage the rover logic, control, and the simulation of the entire communications subsystem.



Terrain

To test if the rover design meets our design objectives the terrain used in the rover model needs to be designed and chosen carefully. As specified by NASA JPL the team has been able to take a scan of a section of the Venus surface using the Java Mission-planning and Analysis Software package (JMARS), we export this terrain as a .stl to be imported into the model. This gives us the capability to test rover traversal on real terrain. The model also uses a specialised terrain file that has been 3D modelled with specific obstacles as a test area, by changing the rover location we can test the rover and ODS in specific scenarios to observe rover behaviour, the test area features a valley, impassable rock field, steep bowl, deep and shallow ditches, and ridge lines. To import the terrains into Simscape, the triangle data from the .stl files is converted into grid point surface data by utilising a script that deletes repeated points and interpolates height data.

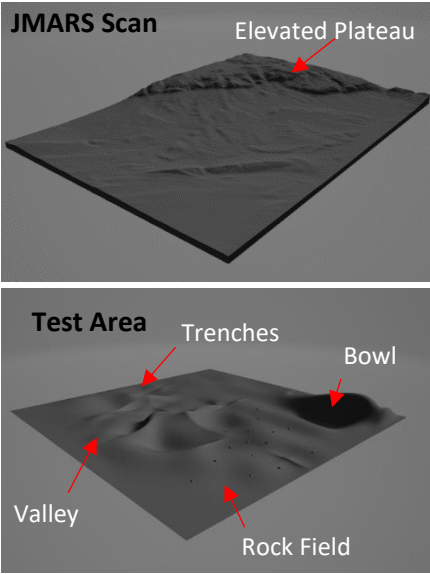
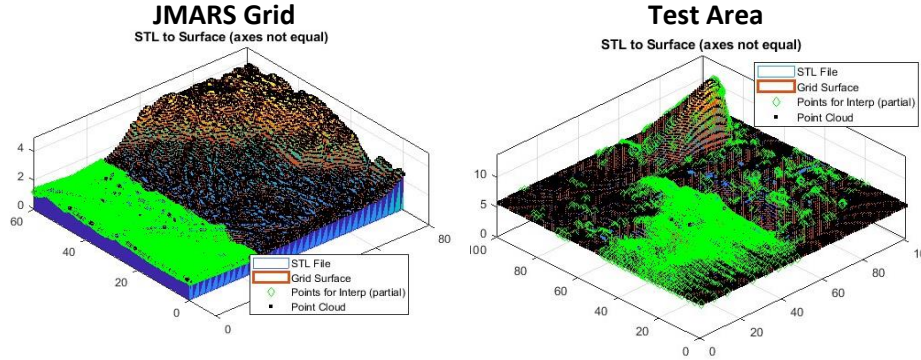


Figure 89: Images showing .stl terrain with main features annotated.

Figure 88: Interpolated grid point data for the JMARS Scan and test Area used in the simulation.

Tracks and Locomotion

Modelling the tracks locomotion system presents several challenges. The rover features a pair of tracks on either side driven of the main body, these are driven by a singular sprocket gear connected directly to the gearbox. Roller bogies act as suspension on the bottom of the rover, and the tracks are tensioned with a pair of rollers on the top and idler gears at the front of the rover. Using Simscape multibody, it is possible to model each track link individually and compute the individual interactions between links, this proved to be too computationally expensive to use for a fast-running model. To improve model performance without impacting results the team needed to create a simplified representation of the track system. Since the tracks directly connects all gears, we can assume that the speed of all the rollers at their outer radius is equal to the speed set by the sprocket gear, we can hence represent the tracks as a series of large rollers driven by this sprocket as seen in figure 92. Each Roller has an associated point cloud object assigned to it (generated by a MATAB script), this is used to model roller contact with the terrain using the spatial force contact block in Simscape. A coefficient of friction of 0.2 is used, this is approximately the value for metal on rock and is chosen [27] to be representative for our simulation. Simscape driveline allows us to drive the sprocket gear using either a torque or angular speed demand through a stiff axil.



Figure 91: Fully segmented track model and simplified track model.

ODS and Chassis

The body of the rover is modelled as a simple extruded trapezoid, details in the chassis are not modelled as only the mass and outer shape need to be represented in the simulation. This acts to improve simulation performance. The simplified tracks are attached to this body, the track links are not modelled so virtual masses are attached to either side of the body to account for the weight and inertia contributions of the track links. ODS arms have been modelled using measurements from the rover CAD files, at the end of each arm the 3 rollers are attached via a pivot, each roller can roll individually and the whole unit can pivot about the ODS arm, as is the case for the actual ODS arms. The arms are attached to pivots on the simplified rover body at the same locations as the actual rover. Contact and clipping with the terrain is handled by assigning each roller an individual point cloud in the same way as the track rollers (figure 94). The live angle of each of the arm relative to the rover body is measured from the joint, this is fed into the ODS Input computer. The wind turbine is also modelled using the proposed design and data from experimental testing.

System Integration

The input and mechanical computer are modelled in Simulink. The ODS arm angle data is passed into the input computer where conditional statements decide if an ODS arm has reached a critical positive or negative angle, if so,

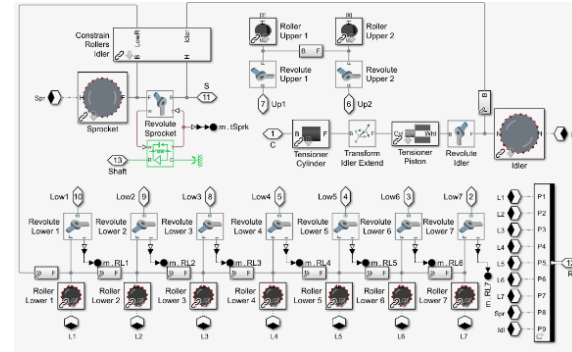


Figure 92: Model of the simplified track

it passes a 1 to the mechanical computer for the respective arm, or a 0 if the arm has not reached a critical angle. The critical angle of each arm is calculated from the arm lengths and the requirement for the rover to scale obstacles 0.35m high. The logic gates in the mechanical computer have been modelled exactly as they are in the actual mechanical computer; hence the outputs of the simulated mechanical computer are identical. Stateflow is an addon package that is used to model instantaneous changes in system state, we use this package to model the rover gearbox, the outputs of the mechanical computer are fed into the Stateflow model. The angular speeds of the track sprocket gears (w_L , w_R) are changed to enable the rover reverse manoeuvres depending on the decision from the mechanical computer. These are outputted back to the Simscape model and control the sprocket speed for the next time step.

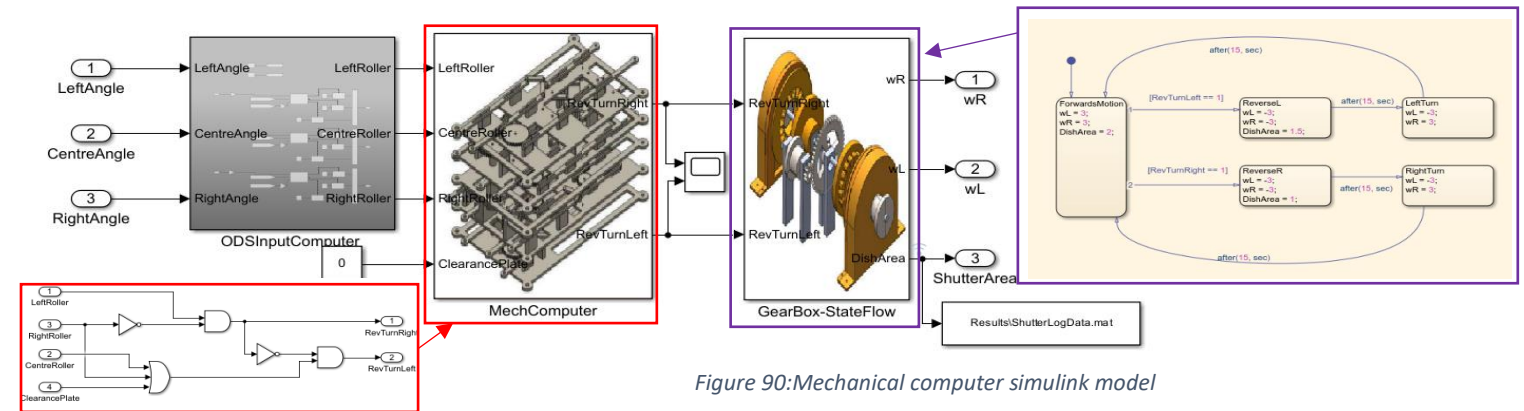


Figure 90: Mechanical computer simulink model

Visual Interface and Results

A user of the model can select either the test area or scanned terrain, they can then position the rover anywhere along the surface in any orientation, and finally can specify a simulation length. These parameters are set in the setup script which will start the simulation when run. When run, the model opens a Mechanics Explorer visual interface in the MATLAB window, this visual interface updates as the simulation runs and can be used to watch the rover traverse the terrain, there are 8 sets of camera angles that the user can choose from. In the mechanics explorer, these camera angles can be set as individual tiles in the window to watch the rover from multiple or more detailed perspectives. The visuals can be paused, sped up, or slowed down even while the simulation is running but of course you can only see up to the latest timestep generated by the model. In addition to the visual interface, all dynamic properties of the rover body (Position, Velocity, Acceleration, Roll, Pitch, Yaw, etc.) are logged and can be access through the Simulink Data Inspector. The logged data and recordings of each camera angle are saved to a new results file per run as .mat and .avi files respectively, this allows the user to compare and share data between runs and to people without access to the model directly.

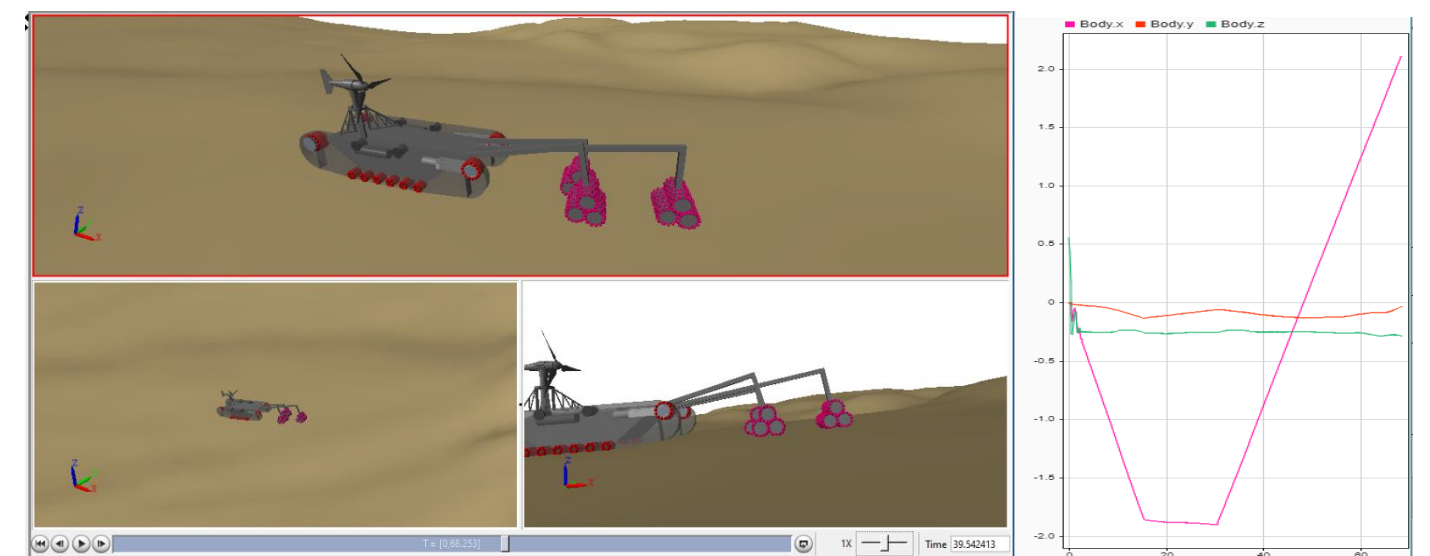


Figure 94: Screenshot of the mechanics explorer window, points clouds for the tracks and rollers clearly shown.. Logged signals of the rover x, y, and z positions are shown on the right.

Communications

Validating the proposed communications system was one of the main objectives of the rover model, the first step was to develop an accurate model of the Venus atmosphere to simulate RF signal losses. Venus-GRAM is a Fortran based program developed by NASA to model estimates of the Venus atmosphere conditions from the surface to 400km altitude. By running and interfacing the data from this program with MATLAB, the team was able to model estimates of temperature, pressure, wind speed, wind direction, density, speed of sound, molar atmosphere make up, average molar mass and 74 other quantities. Combining this data with a series of empirical gas equations derived from a 2019 IEEE Conference paper [28], the team developed a MATLAB script to model the total signal loss due to Atmospheric attenuation/absorption, as well as Free-space path loss, and H2SO4 rain attenuation through the atmosphere. These are functions of signal path distance through the atmosphere in dB/km.

CO2 and N2

$$\alpha_{CN} \approx k_{CN} f^2 p^2 T^{-5},$$

$$k_{CN} = (1.15 \times 10^8 \text{ GHz}^{-2} \text{ K}^{-2} \text{ km}^{-1}) P_{\text{atm}}^{-2} q_{CN};$$

$$q_{CN} = q_{CO2} \left(1 + \frac{q_{CO2} q_{N2}}{4} \right) + 0.0054 q_{N2};$$

SO2

$$\alpha_{SO2} \approx k_{SO2} f^2 p^{1.28} T^{-2.91},$$

$$k_{SO2} = (4.3 \times 10^6 \text{ GHz}^{-2} \text{ K}^{-2.91} \text{ km}^{-1}) P_{\text{atm}}^{-1.28} q_{SO2};$$

H2SO4

$$\alpha_{H2SO4} \approx k_{H2SO4} f^{1.21} p^{1.24} T^{-3},$$

$$k_{H2SO4} = (39.471 \text{ GHz}^{-1.21} (553 \text{ K})^3 \text{ km}^{-1}) P_{\text{atm}}^{-1.24} q_{H2SO4}$$

$$q_{H2SO4} = 10^{-6} \left(\frac{3.7}{1.004 \left(\frac{h-43}{3} \right)^2} + \frac{3.4}{1.09 \left(\frac{h-40}{3} \right)^2} \right) + \frac{3.3}{1.8 \left(\frac{h-39}{3} \right)^2} + \frac{2.4}{1.2 \left(\frac{h-52}{3} \right)^2},$$

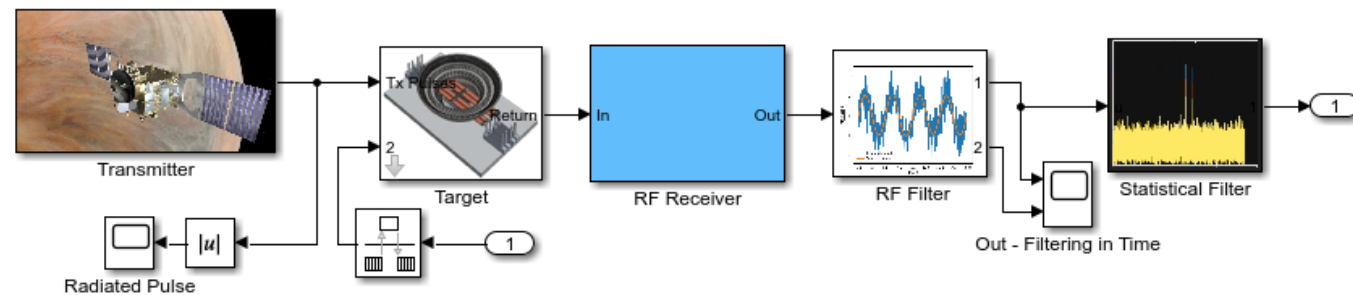


Figure 96: Communications Radar system model, integrated with the Simscape and Simulink Model.

With the signal losses simulated the team developed a radar system simulation using the RF block set, Signal Processing Toolbox, and DSP System Toolbox. The model features a radar pulse generator to generate 260W 8GHz pulses or chirps at a specified cycle rate coming from a satellite in the constellation, these chirps pass through the atmosphere where we apply our calculated atmospheric losses. The pulses are reflected from the communications antenna/dish on the rover, the available area on the dish is changed depending on which shutters are opened and closed, this in turn changes the reflected chirp power. The signal is passed back through the atmosphere where the losses are applied again, they are received by a simulated RF receiver that would be present on the satellite. The returned signal is filtered and is passed through a statistical estimator to estimate which shutters are open on the rover dish based on the returned chirp signal power. The estimator was made by taking large samples (100 seconds of chirps, see figure 98) of returned power and calculating a mean and standard deviation for each shutter state, the returned signals are hence assigned a state if they fall within 1 standard deviation of a mean value (68% certainty). Figure 98 shows multiple overlaid samples of returned chirps for different shutter states. Blue is when all shutters are open; orange when 1 is closed; yellow when 2 are closed. Any less and the signal phases into the noise and becomes undetectable. The rover model changes the shutter state (which shutters are open or closed) based on the rover's actions. When driving forwards, all shutters are open; when the rover performs a manoeuvre, shutters are closed by the gearbox and mechanical computer; when reversing and

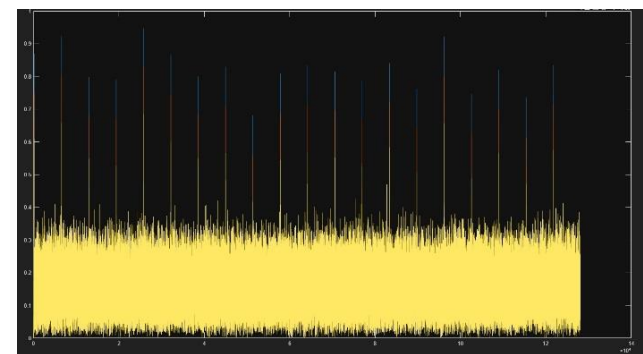


Figure 97: Small plot of returned chirps detected by the simulated satellite. Differing power levels are observable for the different shutter states.

turning left, 1 shutter is closed; when reversing and turning right, 2 shutters are closed. Running a scenario where all manoeuvres are performed over 108 seconds gives us a shutter state output seen in figure 96, which tells us that the rover initially reverses and turns left, drives in forwards motion, and then reverses and turns right. As previously mentioned, the shutter state per time step changes the available area on the dish, figure 96 shows us the reflected chirp power detected by the satellite receiver throughout this scenario. The change in average power level is clearly observable as the shutters are closed and opened. The statistical estimator on the satellite takes this reflected chirp power and estimates how many shutters should be open. The output of the estimator is shown in figure 96. The random nature of noise and its relatively large magnitude to the size of the detections mean that some signals are misidentified and classed into the wrong state by the estimator (only an approximately 68% certainty in correctly classing signals). However, figure 96 shows that since most signals are identified correctly the satellite estimator output follows the actual shutter states on the rover closely. This gives the team confidence and validates that the manoeuvres of the rover can be accurately tracked from orbit using the radar and proposed communications system, in conformation with the COMM4 requirement and the Mission Architecture & Concept of Operations.

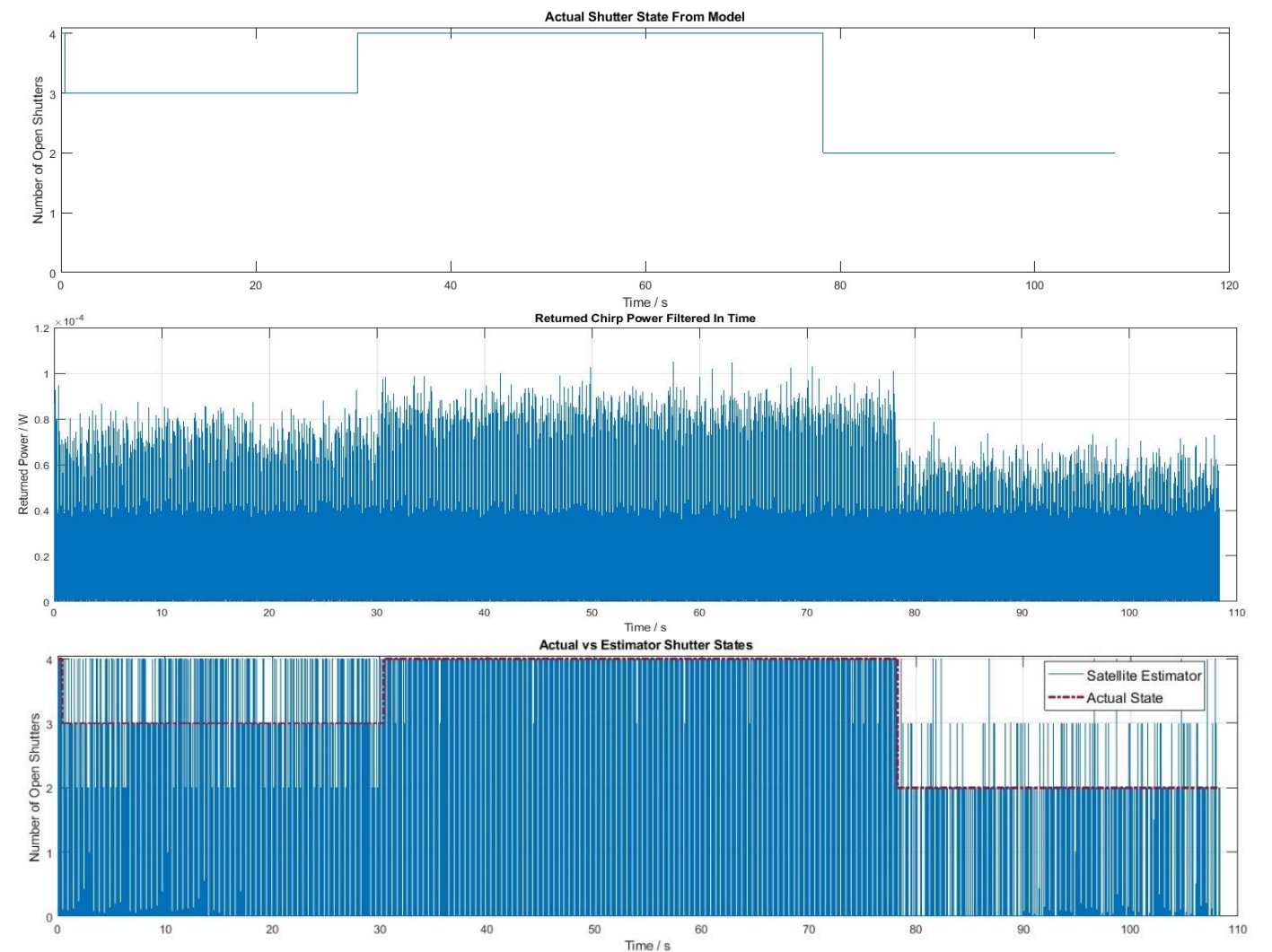


Figure 95: Outputs from the communications subsystem of Actual shutter states from the model, returned chirp power, and estimated states vs actual shutter states.